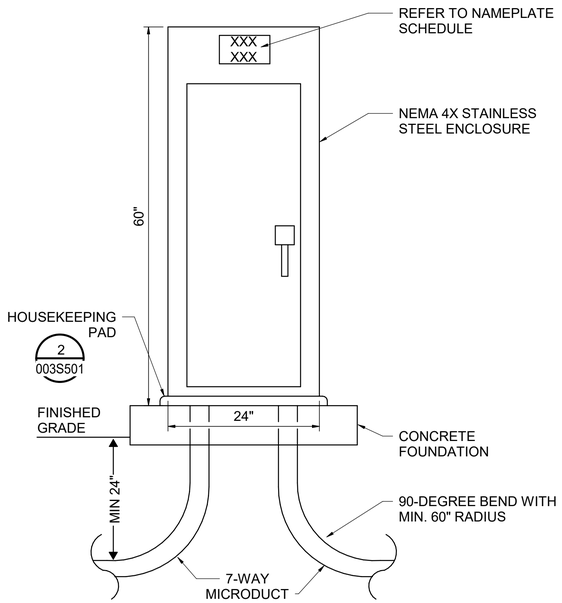
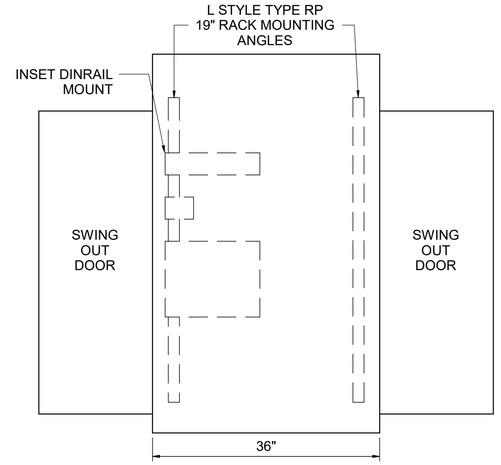


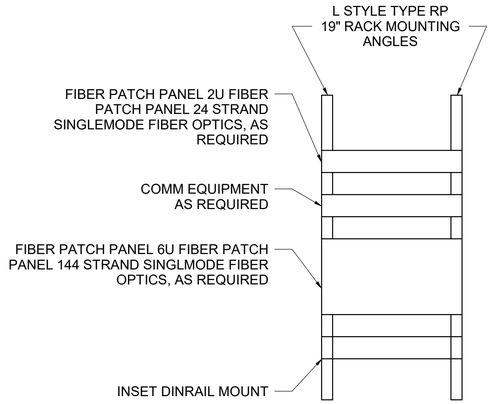
1 009Y501 FIBER OPTIC PATCH PANEL (FOPP##) FRONT EXTERIOR ELEVATION

NOT TO SCALE



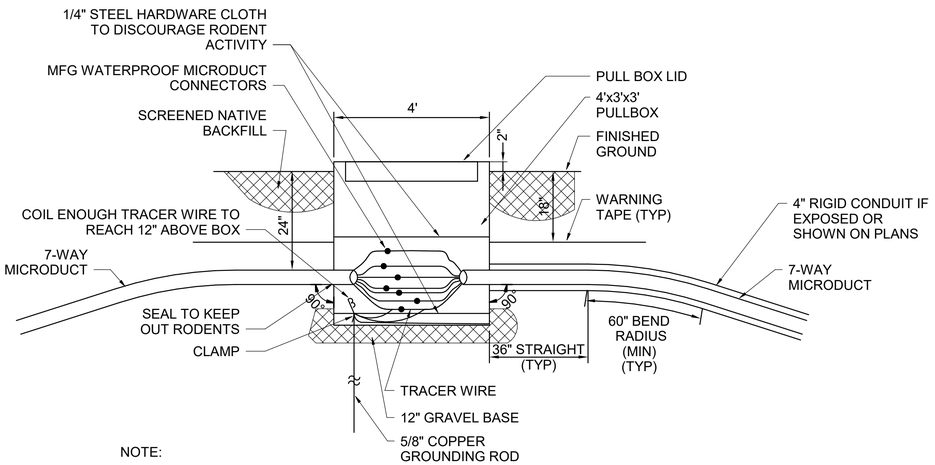
2 009Y501 FIBER OPTIC PATCH PANEL (FOPP##) SIDE EXTERIOR ELEVATION

NOT TO SCALE



3 009Y501 FIBER OPTIC PATCH PANEL (FOPP##) FRONT INTERIOR ELEVATION

NOT TO SCALE



NOTE:

1. SEAL MICRODUCT TO JACKET TO MINIMIZE WATER OR FOREIGN BODY INTRUSION.
2. MATCH INNERDUCT COLOR FOR JUNCTIONS.

4 009Y501 MICRODUCT PULL BOX

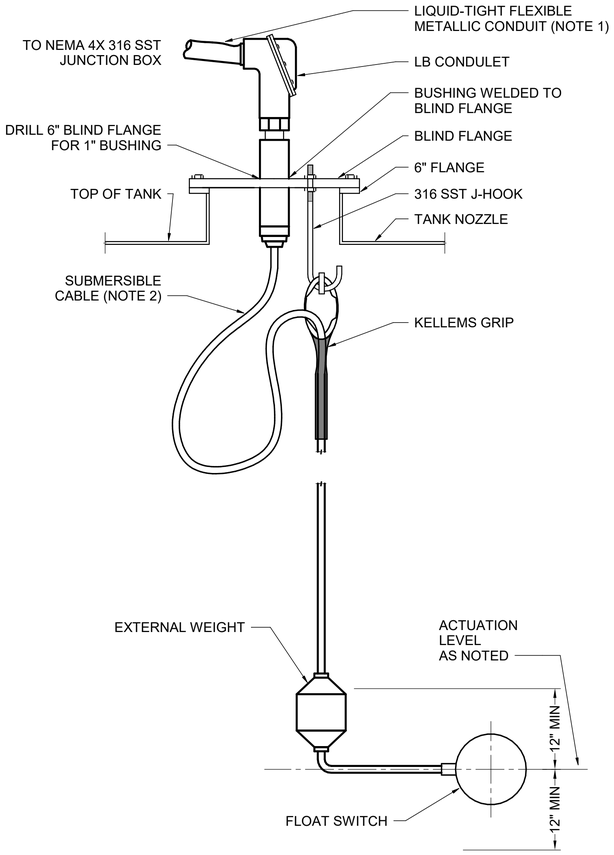
NOT TO SCALE

NAMEPLATE SCHEDULE				
LINE 1	FOPP51	FOPP52	FOPP61	FOPP91
LINE 2	ROARING SPRINGS FIELD OH/UG	ROARING SPRINGS PUMPHOUSE	VISITOR CENTER OH/UG	GENERATOR BUILDING
LINE 3	FIBER OPTIC PATCH PANEL 51	FIBER OPTIC PATCH PANEL 52	FIBER OPTIC PATCH PANEL 61	FIBER OPTIC PATCH PANEL 91

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 04/17/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: P. KEMPF TECH. REVIEW: K CHANDRA DATE: MAY 2026	SUB SHEET NO. 009 Y501	TITLE OF SHEET GENERAL INSTRUMENTATION DETAILS - FIBER NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 89 OF 99

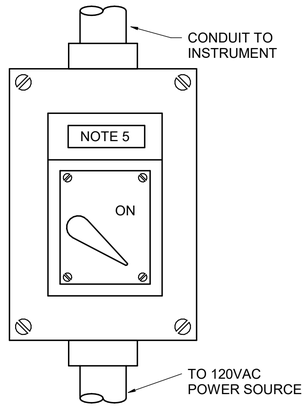
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NOTES:

1. FLEX CONDUIT LENGTH TO BE SUFFICIENT TO ALLOW REMOVAL OF SENSOR.
2. PROVIDE MIN OF 3'-0" SLACK IN CABLE TO ALLOW FOR ADJUSTMENT OF SWITCH ACTUATION LEVEL.

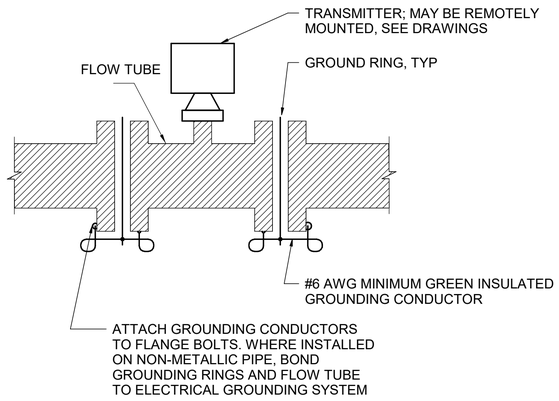
1 / 009Y502 LEVEL FLOAT SWITCH FLANGE MOUNT ON TOP OF TANK
NOT TO SCALE



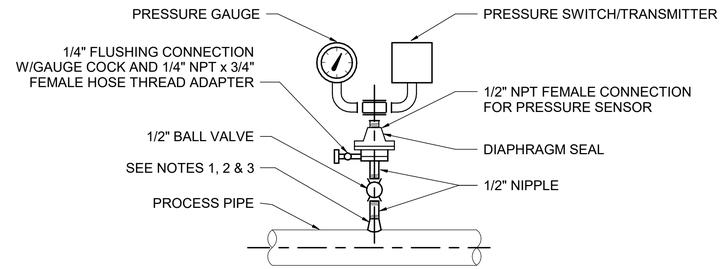
NOTES:

1. SINGLE-PHASE, MANUAL SWITCH REQUIRED FOR EACH AC POWERED FIELD INSTRUMENT, UNLESS OTHERWISE SPECIFIED
2. OPERATING HANDLE PAD-LOCKABLE IN OFF POSITION
3. CAST ALUMINUM NEMA 4X, WATERPROOF ENCLOSURE
4. EATON TYPE MS, OR EQUAL
5. NAME PLATE ATTACHED SECURELY. ENGRAVED WITH INSTRUMENT TAG NUMBER AND DESCRIPTION
6. ENCLOSURES AND COMPONENTS TO BE NEMA 7, NEMA 9 RATED WHERE INSTALLED IN HAZARDOUS RATED LOCATIONS. REFERENCE ELECTRICAL PLAN SHEETS FOR AREA RATINGS.

2 / 009Y502 SINGLE-PHASE CONTROL DISCONNECT
NOT TO SCALE



4 / 009Y502 MAGNETIC TYPE FLOW TRANSMITTER
NOT TO SCALE



NOTES:

1. FOR STEEL, GALVANIZED STEEL AND PVC 2 1/2" AND SMALLER, USE A BUSHING IN A TEE.
2. FOR DUCTILE IRON AND PVC LARGER THAN 2-1/2" ALL SIZES, USE PIPE SADDLE W/ BUSHING. FOR TAPS SHOWN ON FITTINGS USE TAPPED BOSS.
3. FOR STEEL, AND STAINLESS STEEL PIPES 3" AND LARGER, AND PRESSURE VESSELS. USE SHOP INSTALLED THREAD-O-LET, OR EQUAL AS SHOWN.
4. DIAPHRAGM SEAL AND FLUSHING CONNECTION REQUIRED FOR ALL SERVICES EXCEPT AIR, FINISHED WATER, AND FILTRATE.
5. DIAPHRAGM SEAL PRESSURE SWITCH ASSEMBLY TO BE FACTORY ASSEMBLED.
6. TYPICAL FOR INSTALLATIONS WITH EITHER PRESSURE GAUGE, PRESSURE SWITCH/TRANSMITTER OR BOTH.

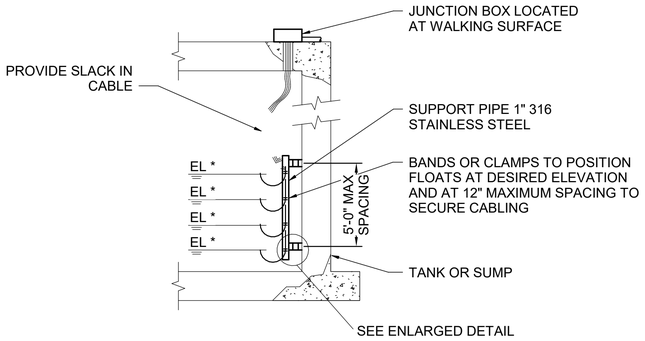
3 / 009Y502 PRESSURE GAUGE AND PRESSURE SWITCH/TRANSMITTER
NOT TO SCALE

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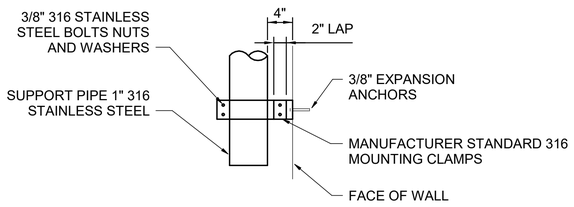
Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 4/27/26 ARIZONA, U.S.A.

A/E FIRM
PRIME:
HDR ENGINEERING, INC.
DENVER, CO 80202

DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
J. BRAINARD	009 Y502	GENERAL INSTRUMENTATION DETAILS - SWITCHES AND TRANSMITTERS	113 182752
DRAWN: P. KEMPF			PMIS/PKG NO. 318719
TECH. REVIEW: K CHANDRA		NORTH RIM - GRAND CANYON NATIONAL PARK	SHEET 90 OF 99
DATE: MAY 2026			



SECTION



ENLARGED DETAIL

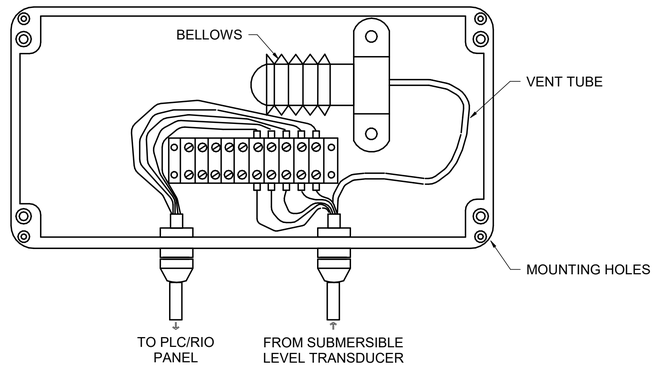
NOTE:

1. WHEN FLOATS SET CLOSER THAN 18" OF LEVEL DIFFERENCE, ROTATE FLOATS AROUND PIPE TO AVOID INTERFERENCE.

* REFER TO SPECIFICATIONS FOR APPROXIMATE MOUNTING HEIGHTS; FIELD ADJUSTMENTS FOR ACTUAL MOUNTING HEIGHTS SHALL BE NECESSARY

1 009Y503 FLOAT SWITCH MOUNTING

NOT TO SCALE

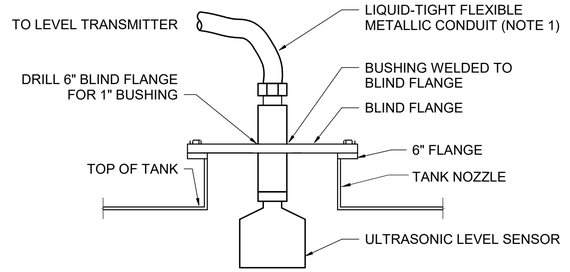


NOTES:

1. PROVIDE NEMA 4X BELLOWS JUNCTION BOX FOR ALL SUBMERSIBLE LEVEL TRANSDUCERS, OR PROVIDE VENT TUBE TERMINATION IN BELLOWS WITHIN ASSOCIATED PLC/RIO PANEL ON BACKPLANE.

2 009Y503 LEVEL TRANSDUCER BELLOWS JUNCTION BOX

NOT TO SCALE

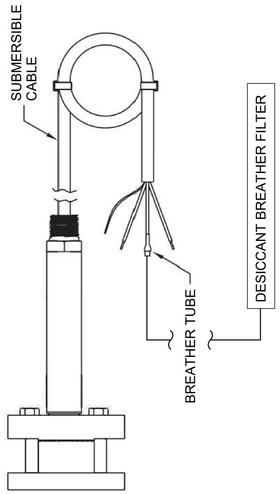


NOTES:

1. FLEX CONDUIT LENGTH TO BE SUFFICIENT TO ALLOW REMOVAL OF SENSOR.
2. USE CONDUIT EXTENSION BETWEEN BUSHING AND SENSOR IF REQUIRED TO ENSURE BOTTOM OF SENSOR IS AT LEAST 1" BELOW TANK TOP.

3 009Y503 ULTRASONIC LEVEL SENSOR FLANGE MOUNT ON TOP OF TANK

NOT TO SCALE

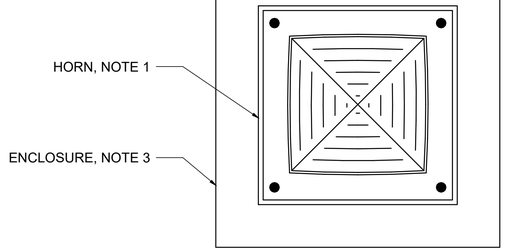


NOTES:

1. JACKETED SHIELDED CABLE WITH ATMOSPHERIC VENT TUBE
2. PLACE DESICANT BREATHER TUBE IN 12x12x6 JUNCTON BOX. SEAL ALL CONDUITS WITH CONDUIT SEAL PUTTY

6 009Y503 HYDROSTATIC LEVEL TRANSMITTER

NOT TO SCALE

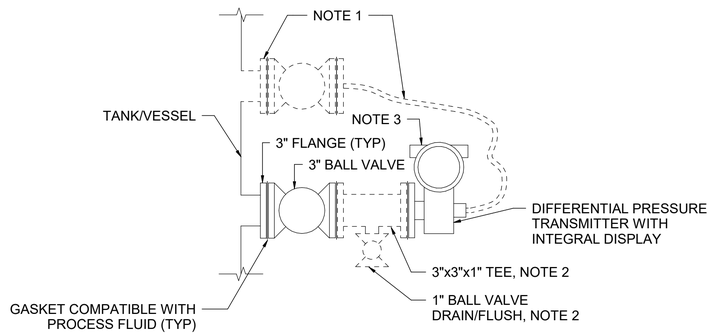


NOTES:

1. INSTALL HORN AND BEACON ACCORDING TO MFR INSTRUCTIONS.
2. CONTRACTOR TO PROVIDE NAMEPLATE TO MATCH TAG NAME SPECIFIC TO EACH SITE.
3. NEMA 7 PANEL TO BE PROVIDED, SIZE AS REQUIRED.

5 009Y503 HORN/BEACON MOUNTING CONFIGURATION

NOT TO SCALE



NOTES:

1. FOR CLOSED VESSEL, POVIDE SECOND FLANGED CONNECTION ABOVE THE MAXIMUM LIQUID LEVEL, AND MANUFACTURER CAPILARY TUBE AND FLANGE CONNECTION.
2. PROVIDE DRAIN/FLUSH VALVE FOR NON-POTABE PROCESS MEDIA.
3. PLUG ALL UNUSED CONDUIT ENTRIES AND PROCESS PORTS.

7 009Y503 LEVEL TRANSMITTER (PRESSURE-BASED) FLANGED SIDE CONNECTION

NOT TO SCALE

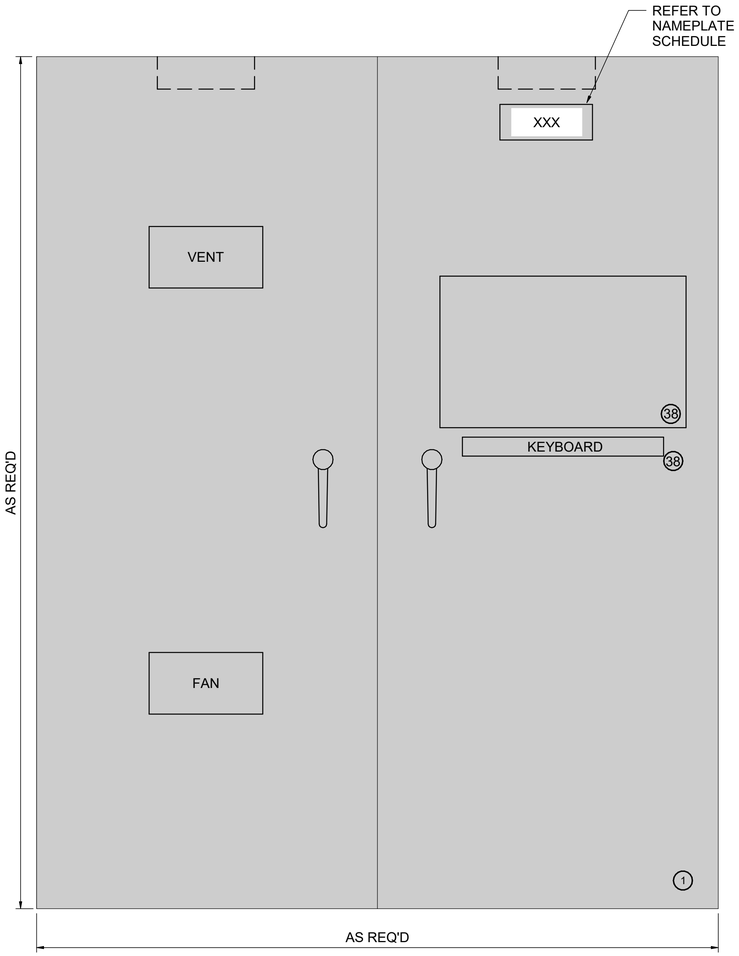


A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: P. KEMPF TECH. REVIEW: K CHANDRA DATE: MAY 2026	SUB SHEET NO. 009 Y503	TITLE OF SHEET GENERAL INSTRUMENTATION DETAILS - GENERAL NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 91 OF 99

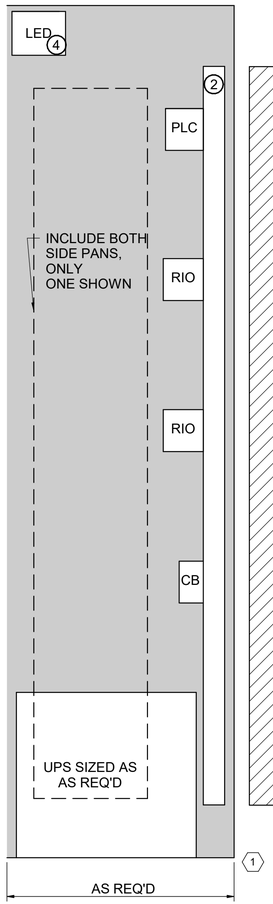
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GENERAL NOTES

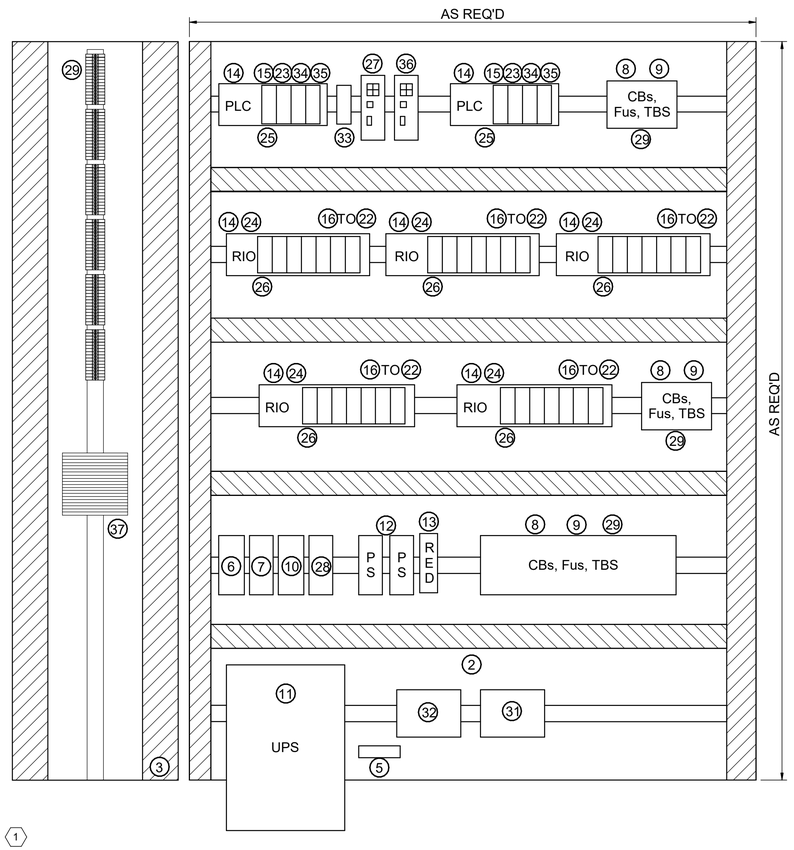
A. PANEL DIMENSIONS VARY BY SITE, REFER TO ELECTRICAL PLAN DRAWINGS.



1 009Y620 FRONT ELEVATION
NOT TO SCALE

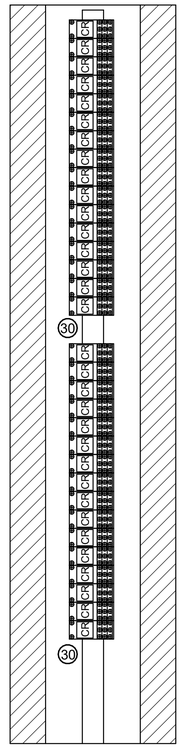


2 009Y620 SIDE EXTERIOR ELEVATION
NOT TO SCALE



3 009Y620 LEFT SIDE INTERIOR
NOT TO SCALE

4 009Y620 INTERIOR ELEVATIONS
NOT TO SCALE



5 009Y620 RIGHT SIDE INTERIOR
NOT TO SCALE

PANEL BOM		
ITEM #	QTY	DESCRIPTION
1	1	NEMA 12 ENCLOSURE
2	1	ENCLOSURE BACKPANEL
3	AS REQ'D	WIREWAY
4	2	LED LIGHT FIXTURE
5	1	GROUND BAR KIT
6	1	SURGE ARRESTER
7	1	MAIN POWER CIRCUIT BREAKER
8	AS REQ'D	CIRCUIT BREAKERS, SIZE AS REQ'D
9	AS REQ'D	FUSES
10	1	CONVENIENCE RECEPTACLE
11	1	UPS
12	2	24 VDC POWER SUPPLY
13	1	REDUNDANCY MODULE
14	AS REQ'D	PLC POWER SUPPLY
15	2	CPU MODULE
16	AS REQ'D	ANALOG INPUT MODULE
17	AS REQ'D	ANALOG OUTPUT MODULE
18	AS REQ'D	DIGITAL INPUT MODULE
19	AS REQ'D	DIGITAL OUTPUT MODULE
20	AS REQ'D	RELAY OUTPUT MODULE
21	AS REQ'D	RTD MODULE
22	AS REQ'D	SPACE FILLER MODULE
23	2	ETHERNET RIO MASTER MODULE
24	AS REQ'D	ETHERNET RIO SLAVE MODULE
25	2	4 SLOT CHASSIS
26	AS REQ'D	9 SLOT CHASSIS
27	1	MANAGED ETHERNET SWITCH
28	1	UPS RECEPTACLE
29	AS REQ'D	TERMINAL BLOCKS
30	AS REQ'D	CONTROL RELAYS
31	1	HEATER W/ T-STAT
32	1	FIBER PATCH PANEL
33	2	DIN RAIL SWITCH
34	2	EITHERNET COMMS MODULE
35	2	DIGITAL INPUT/OUTPUT MODULE
36	1	UNMANAGED ETHERNET SWITCH
37	AS REQ'D	SIGNAL ISOLATORS
38	AS REQ'D	TOUCHSCREEN W/ WALL MOUNT KEYBOARD

NAMEPLATE SCHEDULE		
LINE 1	ROARING SPRINGS	NORTH RIM
LINE 2	PUMP HOUSE	WATER TREATMENT PLANT
LINE 3	PLC CONTROL PANEL	PLC CONTROL PANEL
ABBREVIATION	RSPH PLC CP	NR WTP PLC CP

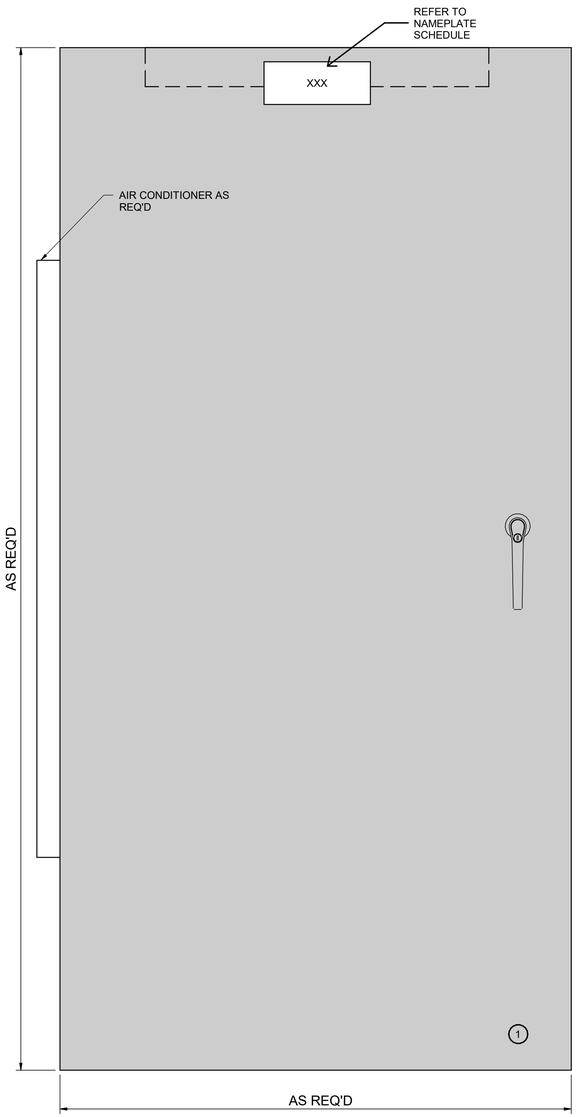
Registered Professional Engineer (Certified) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 3/17/26 ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: S. BILD TECH. REVIEW: K CHANDRA DATE: MAY 2026	SUB SHEET NO. 009 Y620	TITLE OF SHEET GENERAL TYPICAL CONTROL PANEL LAYOUT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 92 OF 99

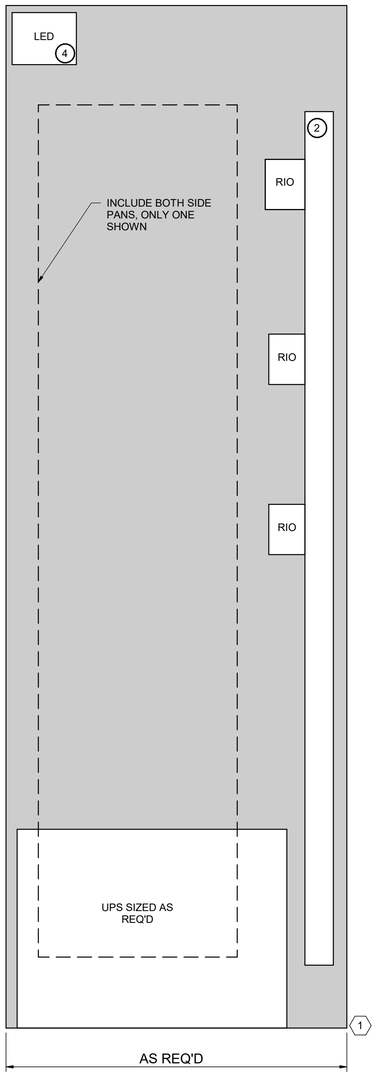
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KEYNOTES: ⟨X⟩

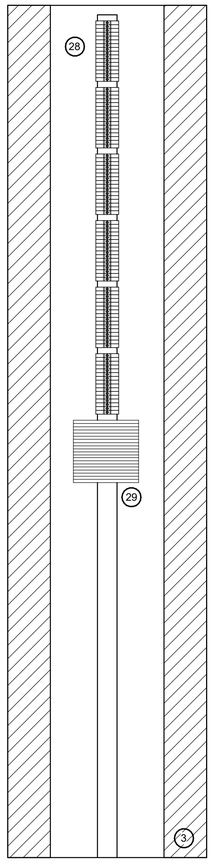
1. PANEL DIMENSIONS VARY BY SITE, REFER TO ELECTRICAL PLAN DRAWINGS.



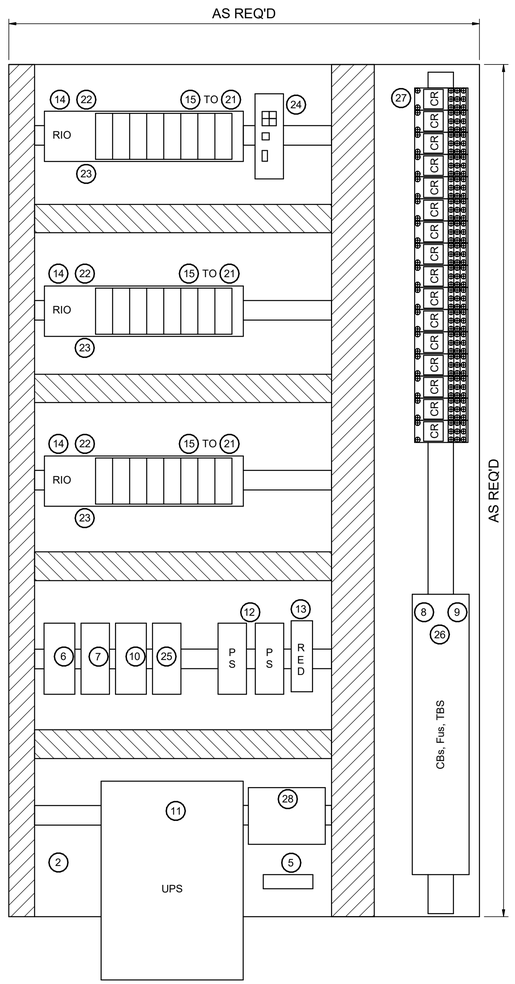
1 FRONT ELEVATION
009Y621
NOT TO SCALE



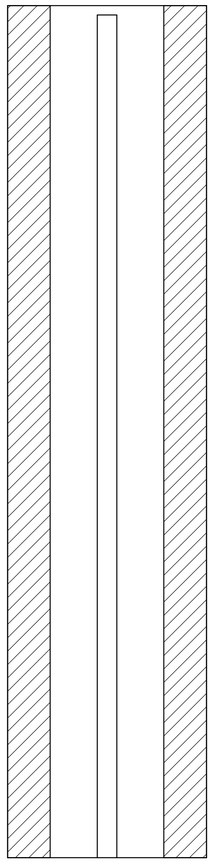
2 SIDE EXTERIOR ELEVATION
009Y621
NOT TO SCALE



3 LEFT SIDE INTERIOR
009Y621
NOT TO SCALE



4 INTERIOR ELEVATIONS
009Y621
NOT TO SCALE



5 RIGHT SIDE INTERIOR
009Y621
NOT TO SCALE

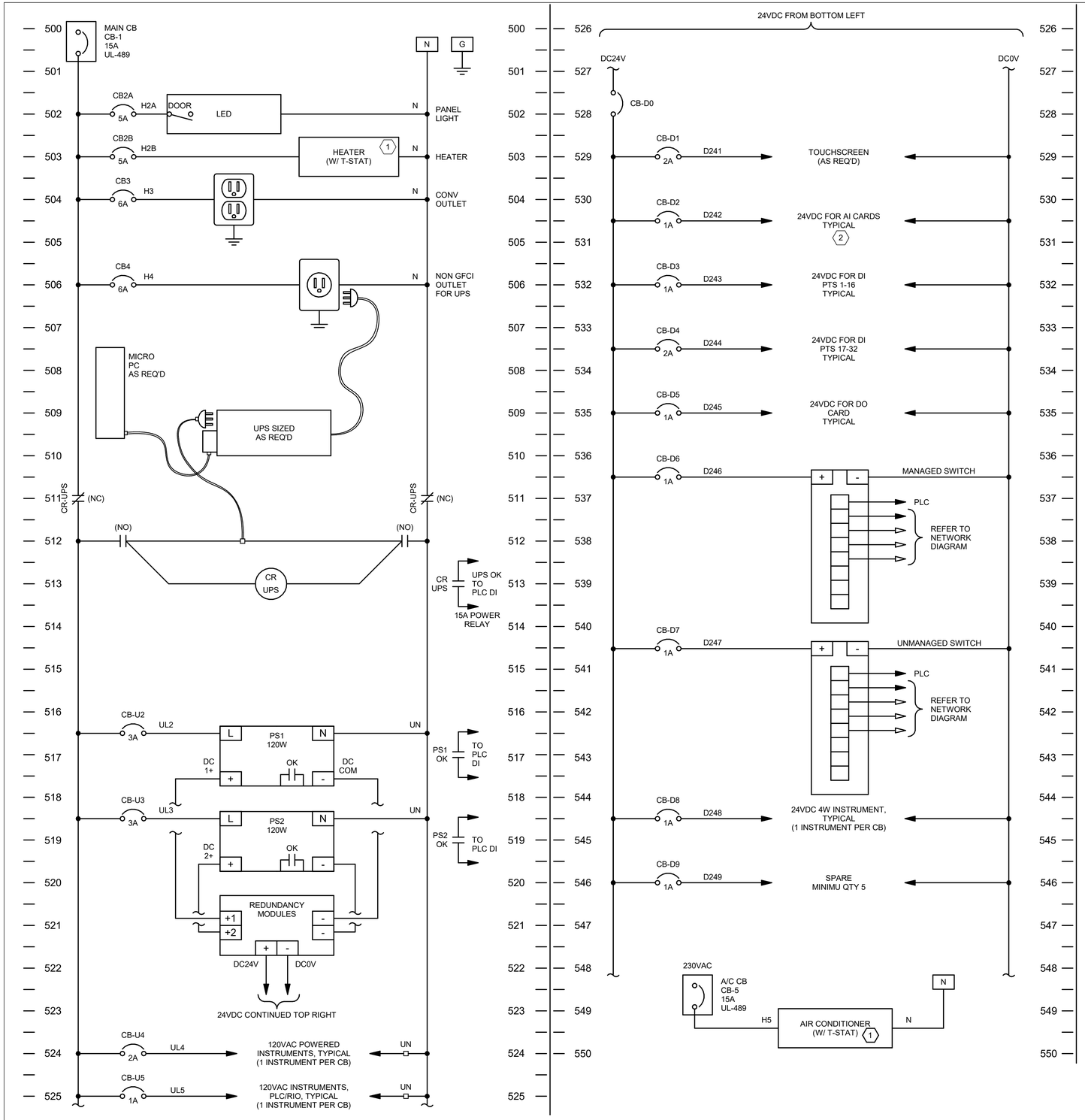
RIO PANEL BOM		
ITEM #	QTY	DESCRIPTION
1	1	NEMA 12 ENCLOSURE
2	1	ENCLOSURE BACKPANEL
3	AS REQ'D	WIREWAY
4	2	LED LIGHT FIXTURE
5	1	GROUND BAR KIT
6	1	SURGE ARRESTER
7	1	MAIN POWER CIRCUIT BREAKER
8	AS REQ'D	CIRCUIT BREAKERS, SIZE AS REQ'D
9	AS REQ'D	FUSES
10	1	CONVENIENCE RECEPTACLE
11	1	UPS
12	2	24 VDC POWER SUPPLY
13	1	REDUNDANCY MODULE
14	AS REQ'D	PLC POWER SUPPLY
15	AS REQ'D	ANALOG INPUT MODULE
16	AS REQ'D	ANALOG OUTPUT MODULE
17	AS REQ'D	DIGITAL INPUT MODULE
18	AS REQ'D	DIGITAL OUTPUT MODULE
19	AS REQ'D	RELAY OUTPUT MODULE
20	AS REQ'D	RTD MODULE
21	AS REQ'D	SPACE FILLER MODULE
22	AS REQ'D	ETHERNET RIO SLAVE MODULE
23	AS REQ'D	9 SLOT CHASSIS
24	1	UNMANAGED ETHERNET SWITCH
25	2	UPS RECEPTACLE
26	AS REQ'D	TERMINAL BLOCKS
27	AS REQ'D	CONTROL RELAYS
28	1	HEATER W/ T-STAT
29	AS REQ'D	SIGNAL ISOLATORS

NAMEPLATE SCHEDULE	
LINE 1	NORTH RIM
LINE 2	WATER TREATMENT PLANT
LINE 3	CHEMICAL RIO CONTROL PANEL
ABBREVIATION	NR WTP CHEMICAL RIO CP

Registered Professional Engineer (Inactive)
CERTIFICATE NO.
85305
JOSEPHINE A BRAINARD
Date Signed 11/1/99
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: P. KEMPF TECH. REVIEW: K CHANDRA DATE: MAY 2026	SUB SHEET NO. 009 Y621	TITLE OF SHEET GENERAL TYPICAL RIO PANEL LAYOUT NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 93 OF 99

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TYPICAL CONTROL PANEL POWER SCHEMATIC

GENERAL NOTES

A. SCHEMATIC IS DIAGRAMMATIC. ACTUAL DEVICES & WIRES TO BE INSTALLED PER MANUFACTURERS INSTRUCTIONS.

KEYNOTES: (X)

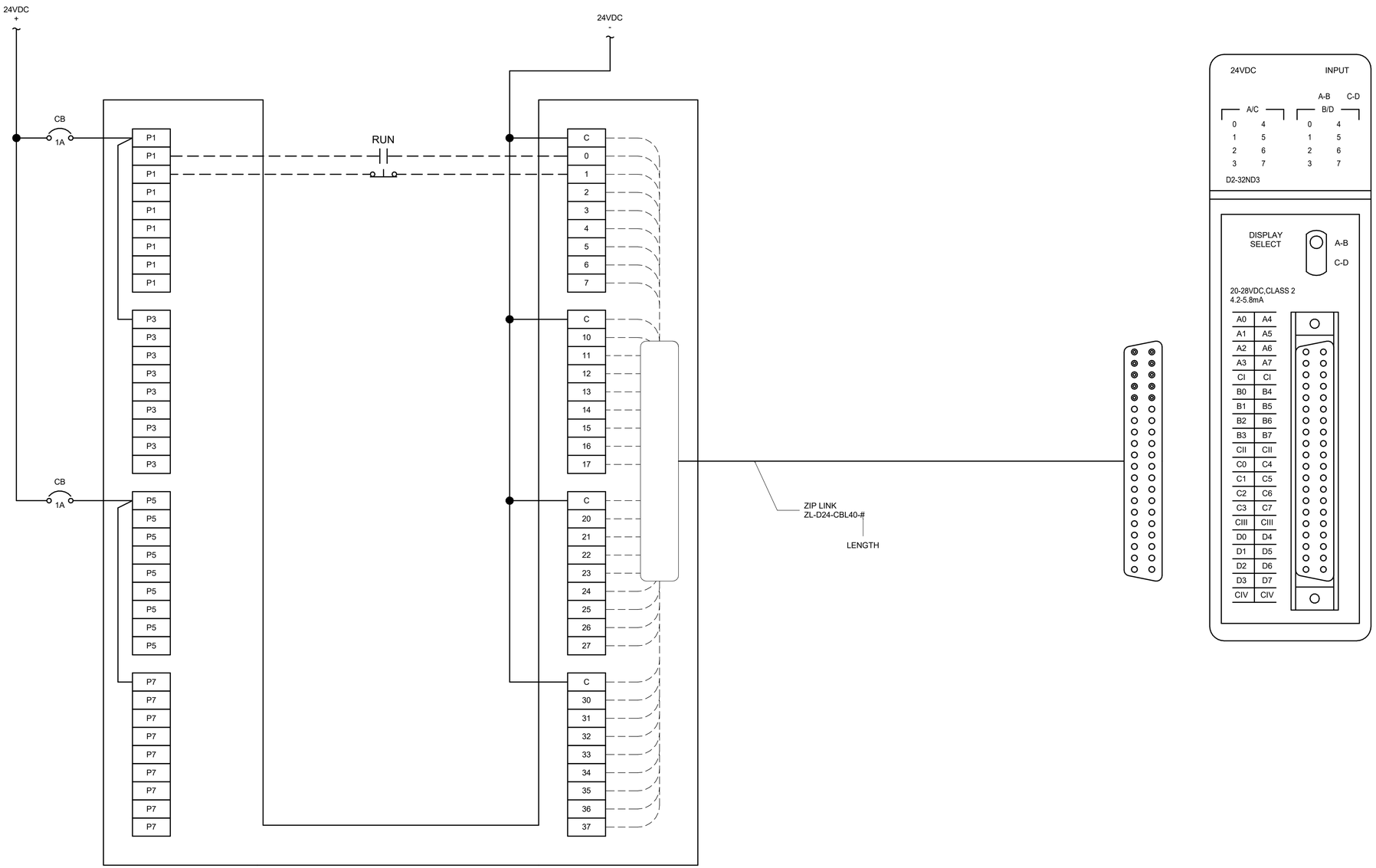
1 PROVIDE PER HEAT DISSIPATION CALCULATIONS AS REQUIRED.

2 PROVIDE BUT DO NOT CONNECT PANEL POWER FOR AI CARDS CONTAINING WEIGHT TRANSMITTER SIGNALS. SEE SPECIFICATION 40 61 93 PROCESS CONTROL SYSTEM INPUT/OUTPUT LIST.

Registered Professional Engineer (Electrical)
CERTIFICATE NO.
85305
JOSEPHINE A BRAINARD
Date Signed 04/17/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD CADD N. LYNCH TECH. REVIEW: K CHANDRA DATE: MAY 2026	SUB SHEET NO. 009 Y622	TITLE OF SHEET GENERAL TYPICAL CONTROL PANEL POWER SCHEMATIC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 94 OF 99

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TYPICAL DIGITAL INPUT MODULE WIRING

GENERAL NOTES

A. WIRE ALL POINTS OUT AS SHOWN INCLUDING SPARES & FUTURE POINTS.

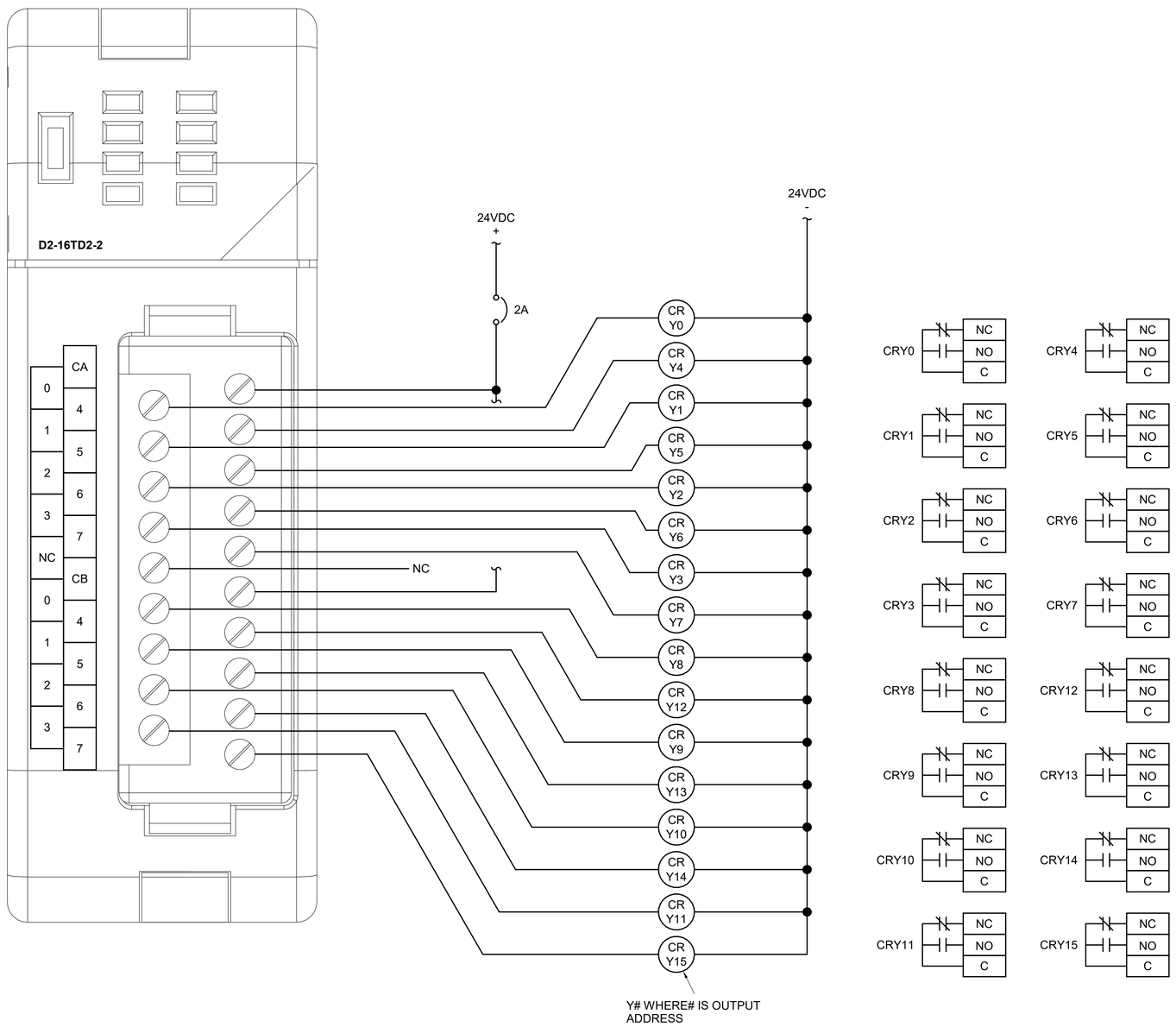
B. WIRING SHOWN IS DIAGRAMMATIC. ACTUAL DEVICES AND WIRES TO BE INSTALLED PER MFG INSTRUCTIONS.

C. I/O PER SPECIFICATION 40 61 93 PROCESS CONTROL SYSTEM INPUT-OUTPUT LIST.

Registered Professional Engineer Certificate No. 85305 JOSEPHINE A BRAINARD Date Signed 4/27/26 ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: D. LOPEZ TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 009 Y630	TITLE OF SHEET GENERAL TYPICAL DI WIRING SCHEMATIC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 95 OF 99

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TYPICAL RELAY OUTPUT MODULE WIRING

GENERAL NOTES

A. WIRE ALL POINTS OUT AS SHOWN INCLUDING SPARES & FUTURE POINTS.

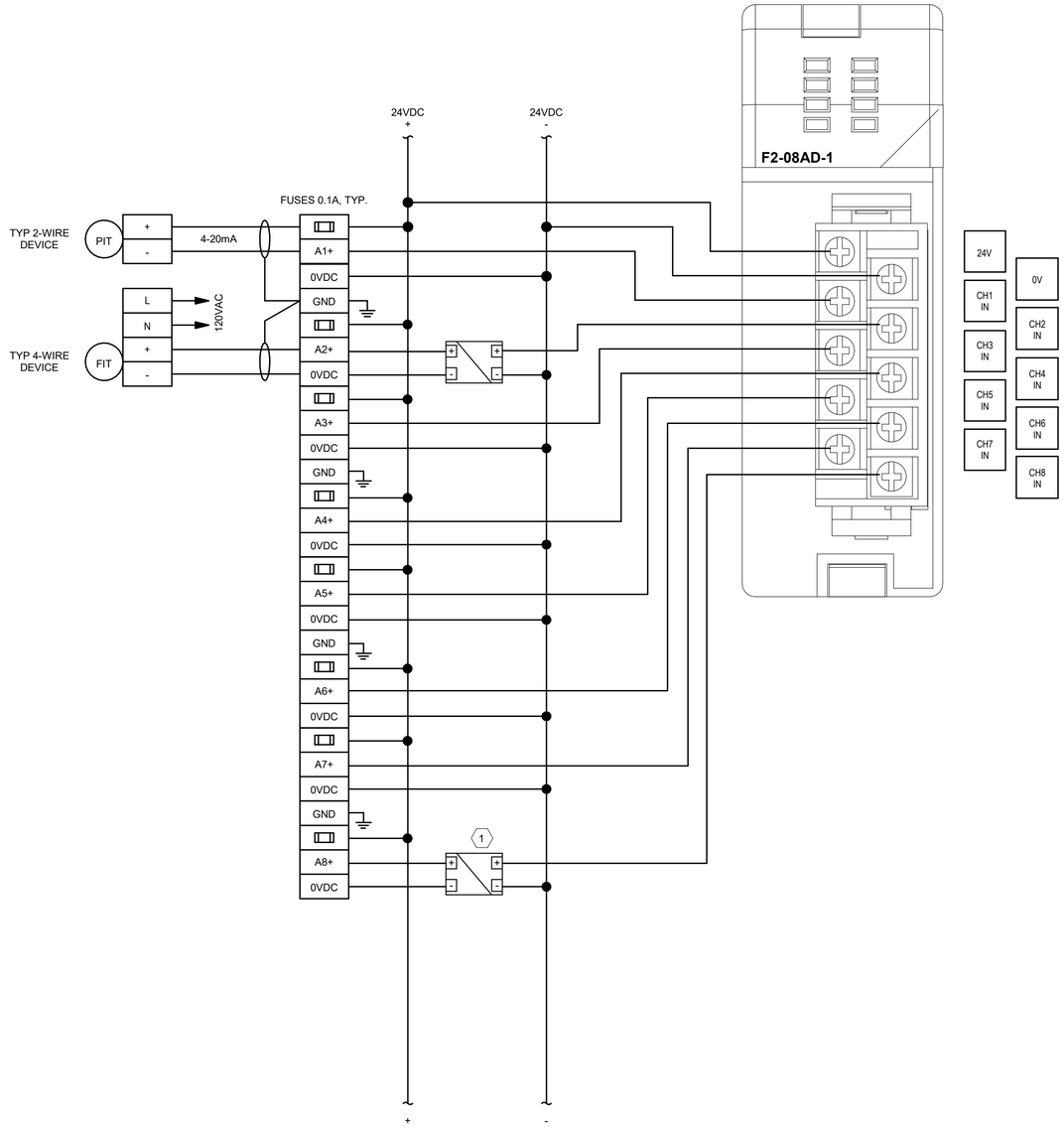
B. WIRING SHOWN IS DIAGRAMMATIC. ACTUAL DEVICES AND WIRES TO BE INSTALLED PER MFG INSTRUCTIONS.

C. I/O PER SPECIFICATION 40 61 93 PROCESS CONTROL SYSTEM INPUT-OUTPUT LIST.

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Registered Professional Engineer (Certified)
CERTIFICATE NO.
85305
JOSEPHINE A BRAINARD
Date Signed 4/1/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: D. LOPEZ TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 009 Y631	TITLE OF SHEET GENERAL TYPICAL DO WIRING SCHEMATIC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 96 OF 99



TYPICAL 4-20mA ANALOG INPUT MODULE WIRING

GENERAL NOTES

A. WIRE ALL POINTS OUT AS SHOWN INCLUDING SPARES & FUTURE POINTS.

B. WIRING SHOWN IS DIAGRAMMATIC. ACTUAL DEVICES AND WIRES TO BE INSTALLED PER MFG INSTRUCTIONS.

C. I/O PER SPECIFICATION 40 61 93 PROCESS CONTROL SYSTEM INPUT-OUTPUT LIST.

KEY NOTES (X)

1. PROVIDE SIGNAL ISOLATORS FOR A MINIMUM OF 25% SPARE AI POINTS.

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Registered Professional Engineer (Electrical)
CERTIFICATE NO.
85305
JOSEPHINE A BRAINARD
Date Signed 5/1/26
ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: D. LOPEZ TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 009 Y632	TITLE OF SHEET GENERAL TYPICAL AI WIRING SCHEMATIC NORTH RIM - GRAND CANYON NATIONAL PARK

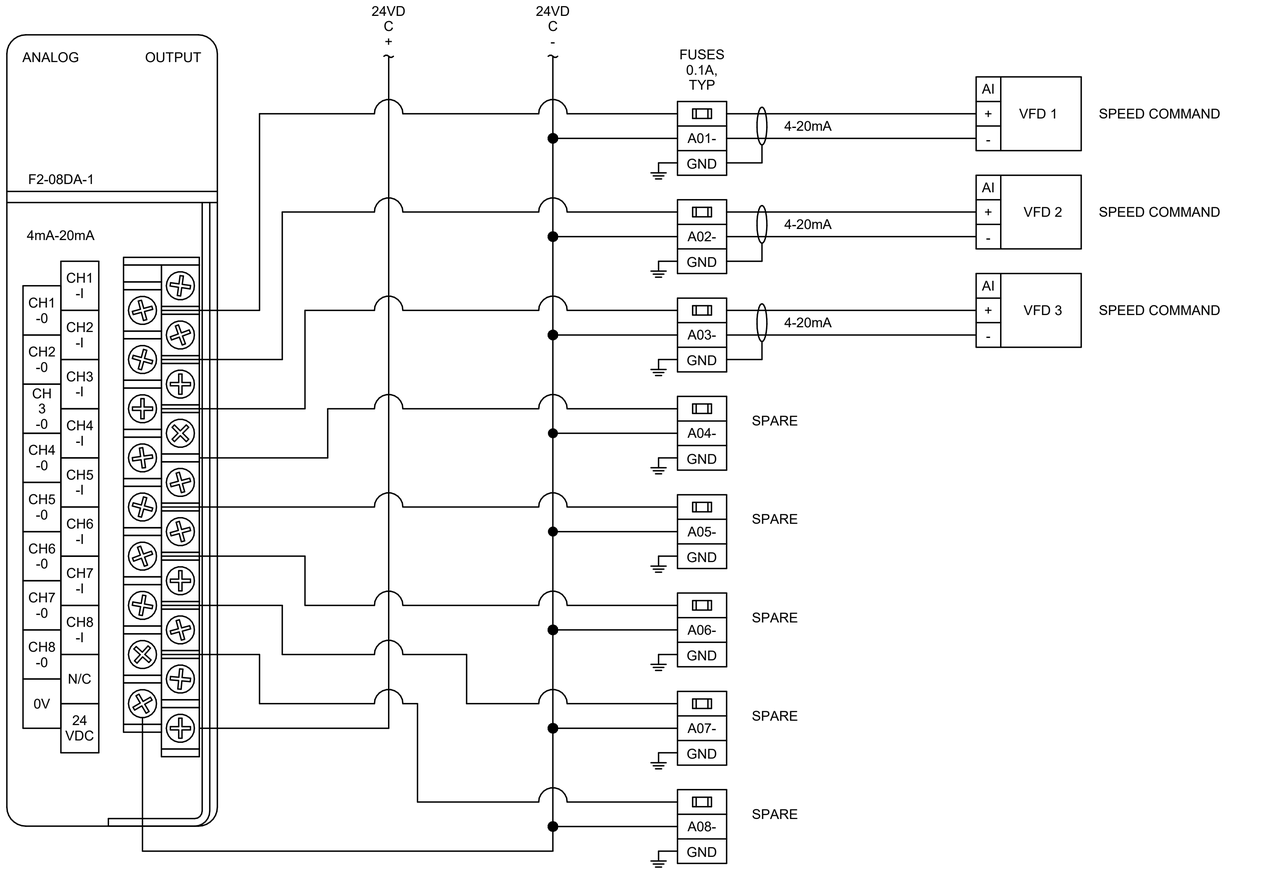
DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	97 OF 99

GENERAL NOTES

A. WIRE ALL POINTS OUT AS SHOWN INCLUDING SPARES & FUTURE POINTS.

B. WIRING SHOWN IS DIAGRAMMATIC. ACTUAL DEVICES AND WIRES TO BE INSTALLED PER MFG INSTRUCTIONS.

C. I/O PER SPECIFICATION 40 61 93 PROCESS CONTROL SYSTEM INPUT-OUTPUT LIST.



TYPICAL ANALOG OUTPUT MODULE WIRING

Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 4/1/26 ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	J. BRAINARD DRAWN: N. LYNCH TECH. REVIEW: K. CHANDRA DATE: MAY 2026	009 Y633	GENERAL TYPICAL AO WIRING SCHEMATIC NORTH RIM - GRAND CANYON NATIONAL PARK	113 182752 PMIS/PKG NO. 318719 SHEET 98 OF 99

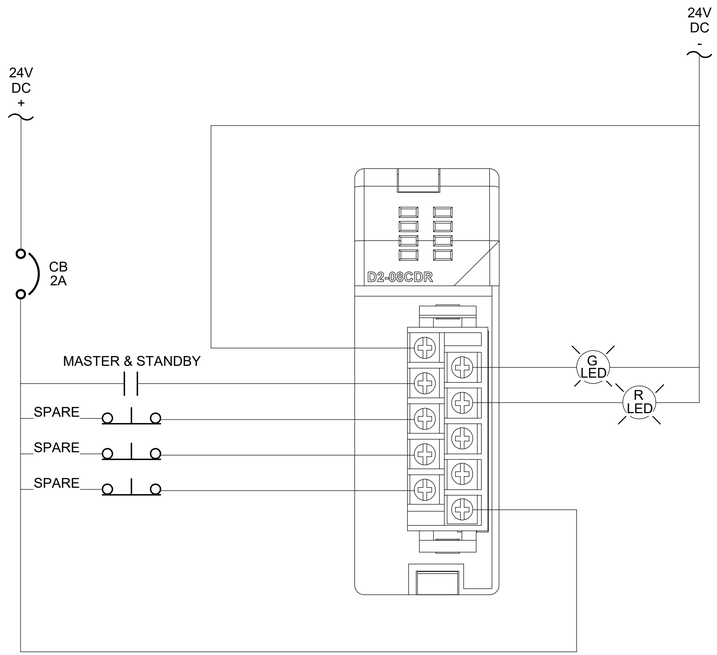
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GENERAL NOTES

A. WIRE ALL POINTS OUT AS SHOWN INCLUDING SPARES & FUTURE POINTS.

B. WIRING SHOWN IS DIAGRAMMATIC. ACTUAL DEVICES AND WIRES TO BE INSTALLED PER MFG INSTRUCTIONS.

C. I/O PER SPECIFICATION 40 61 93 PROCESS CONTROL SYSTEM INPUT-OUTPUT LIST.



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Registered Professional Engineer (Electrical)
CERTIFICATE NO.
85305
JOSEPHINE A BRAINARD
Date Signed 04/27/26
ARIZONA, U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: J. BRAINARD DRAWN: N. LYNCH TECH. REVIEW: K. CHANDRA DATE: MAY 2026	SUB SHEET NO. 009 Y634	TITLE OF SHEET GENERAL TYPICAL INPUT/RELAY WIRING SCHEMATIC NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 99 OF 99